Yugansh Sharma

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Objective

Final-year CSE student specializing in backend engineering (**Java**, **Spring Boot**, **Microservices**) and applied **GenAI**, with experience integrating **LLMs** into production-style applications. Solved 400+ DSA problems. Focused on building **AI-powered backend systems** that are scalable, reliable, and production-ready.

Technical Skills

Programming Languages: Java, SQL
Web: HTML, CSS
Backend: Spring Boot, Spring Cloud, Spring Security, Keycloak, Apache Kafka, REST APIs, Microservices
GenAI: Spring AI, Gemini API, RAG, Vector Embeddings, Vector Databases, AI Agents & Tool Calling, Model Context Protocol (MCP), Prompt Engineering, Streaming Responses (SSE)
Databases: PostgreSQL, MongoDB, MySQL, Redis
Core CS: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), DBMS
Tools: Git, GitHub, Maven, Docker, Postman

Projects

FitPilot: AI-Powered Fitness Tracker | *Spring Boot, Keycloak, Kafka, MongoDB, Gemini API* GitHub

- Built a **microservices-based AI** fitness platform (User, Activity, AI services) that generates personalized workout analysis, improvement areas, and safety guidelines for each logged activity using the **Google Gemini API**.
- Secured the system with **Keycloak (OAuth2/JWT)** enforced at a Spring Cloud Gateway, which also auto-syncs authenticated users into the User service on first login.
- Implemented event-driven communication using **Apache Kafka** to decouple Activity and AI services, alongside synchronous REST calls (via Eureka-based load-balanced **WebClient**) for user validation.
- Applied polyglot persistence (**PostgreSQL** for users, **MongoDB** for activities and recommendations) and centralized configuration with **Spring Cloud Config**.

AetherLink: URL Shortener & Analytics Platform | *Spring Boot · PostgreSQL · Docker · JWT*

- Implemented stateless **JWT authentication** with a custom **Spring Security** authentication filter, **BCrypt password hashing**, and **role-based access control** for protected **REST endpoints**.
- Designed a normalized **User–UrlMapping–ClickEvent** schema with **Spring Data JPA** and **PostgreSQL** on **Neon**, enabling click tracking and date-based analytics.
- Generated **8-character alphanumeric slugs** with **HTTP 302 redirects**, supporting path-based and subdomain-based short-link routing; containerized the backend with **Docker** and published images to **Docker Hub**.
- Deployed the backend and database using **Render** and **Neon PostgreSQL**, with a custom domain, DNS configuration, and automatic HTTPS.

AI Agent: Website Generator & Tool-Calling API | *Java 21, Spring Boot, Spring AI, Gemini API* GitHub

- Built an autonomous **AI agent** using **Spring AI** `ChatClient` and **LLM tool calling** that generates complete multi-file websites (HTML/CSS/JS) from prompts using a plan → write → verify → fix loop with retry on tool errors.
- Designed a **sandboxed file-system toolset** (`@Tool`: create, write, read, list) with path normalization and workspace-boundary checks to prevent **path traversal** by the model.
- Exposed a **streaming chat API** (**SSE**, Reactor `Flux`) with multi-turn context and `@Tool`-based integrations: arithmetic calculator, live **WeatherAPI** and **Frankfurter** currency-exchange lookups via **RestClient**.
- Kept the LLM provider-agnostic: **Gemini** via an OpenAI-compatible endpoint, with model, base URL and API keys externalized as environment variables.

Achievements

- Solved **400+ Data Structures and Algorithms (DSA) problems** across LeetCode, CodeChef, HackerRank and GeeksforGeeks (Arrays, Strings, Linked Lists, Stacks, Queues, Trees); profile: codolio.com/Yugansh.

Education

Noida Institute of Engineering and Technology Greater Noida, India
Bachelor of Technology (B.Tech), Computer Science and Engineering | **CGPA: 7.2/10** 2022 - 2026